Intervention: Finding Intercepts Name: ____________________

The Golden Rules of Intercepts

1. The y-intercept (Start Value)

- Where the graph crosses the vertical line.
- **Rule:** Set $\mathbf{x = 0}$ and simplify.

2. The x-intercept (Solution)

- Where the graph hits the ground (height is 0).
- **Rule:** Set $\mathbf{y = 0}$ and solve for x.

Type A: Cubic $(\dots)^3$

To solve for x, isolate the parenthesis and take the **Cube Root**.

Example: $y = 2(x-1)^3 + 16$

1. Find y-int (Set $x = 0$):

$$y = 2(\mathbf{0} - 1)^3 + 16$$
$$y = 2(-1)^3 + 16$$
$$y = -2 + 16 \quad \rightarrow \quad \mathbf{y = 14}$$

2. Find x-int (Set $y = 0$):

$$0 = 2(x-1)^3 + 16$$
$$-16 = 2(x-1)^3$$
$$-8 = (x-1)^3$$
$$\sqrt[3]{-8} = x - 1 \quad \leftarrow \textbf{ROOT IT!}$$
$$-2 = x - 1$$
$$\mathbf{-1 = x}$$

Type B: Cube Root $\sqrt[3]{\dots}$

To solve for x, isolate the root and raise to the **Power of 3**.

Example: $y = \sqrt[3]{x-5} + 2$

1. Find y-int (Set $x = 0$):

$$y = \sqrt[3]{\mathbf{0} - 5} + 2$$
$$y \approx -1.7 + 2$$
$$\mathbf{y \approx 0.3} \quad \text{(decimals are ok)}$$

2. Find x-int (Set $y = 0$):

$$0 = \sqrt[3]{x-5} + 2$$
$$-2 = \sqrt[3]{x-5}$$
$$(-2)^3 = (\sqrt[3]{x-5})^3 \quad \leftarrow \textbf{CUBE IT!}$$
$$-8 = x - 5$$
$$+5 \qquad +5$$
$$\mathbf{-3 = x}$$

The "Un-doing" Chart:

If you see a **Cubic** $(\dots)^3$ $\xrightarrow{\text{undo it with}}$ **Cube Root** $\sqrt[3]{\dots}$

If you see a **Cube Root** $\sqrt[3]{\dots}$ $\xrightarrow{\text{undo it with}}$ **Power of 3** $(\dots)^3$

Practice: Finding Intercepts

Directions: Determine if the function is Cubic or Cube Root, then find the x and y intercepts.

1. Cubic Function: $f(x) = (x-4)^3 - 8$

A) Find y-intercept (Set $x = 0$)

Answer: (,)

B) Find x-intercept (Set $y = 0$)

Answer: (,)

2. Cube Root Function: $f(x) = \sqrt[3]{x+4} - 2$

A) Find y-intercept (Set $x = 0$)

Answer: (,)

B) Find x-intercept (Set $y = 0$)

Answer: (,)

3. Challenge: $f(x) = -2(x+1)^3 + 54$

A) Find y-intercept ($x = 0$)

Answer: (,)

B) Find x-intercept ($y = 0$)

Answer: (,)